

DAVID W. KASTNER

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Education

Ph.D. Bioengineering	MIT, Cambridge, MA	2025
	GPA 5.0 Graduate Student Spotlight	
B.S. Biophysics	BYU, Provo, UT	2019
	GPA 4.0 Honors <i>Summa Cum Laude</i> , Valedictorian Speech	

Industry Experience

Pfizer Pharmaceutical Internship – Cambridge, MA Sept 2023 – Sept 2024

Machine Learning Computational Sciences (MLCS) Group

- Integrated modern ML models into the CADD workflow for Pfizer's molecular glue initiative
- Developed ML and physics-based workflows for predicting molecular-glue degrader binding
- Benchmarked generative models AF2, Boltz-1, DiffDock-L, RFAA, Chai-1, and Protenix for glues
- 50+ billion molecule structure-based high-throughput virtual screen of molecular glues

Professional CADD and Leadership Certifications

[Molecular Modeling in Drug Discovery](#) issued by Schrödinger

- LigPrep, Epik, Glide, and Ligand Designer in Maestro; ADMET profiling in LiveDesign Free
- [Free Energy Calculations for Drug Design with FEP+](#) issued by Schrödinger
- REST sampling, λ windows, core-matching, and FEP-graph building with Maestro and Desmond.
- [Fundamentals of Deep Learning](#) issued by NVIDIA
- PyTorch, CNNs, data augmentation, transfer learning, and natural language processing
- [Fundamentals of Accelerated Computing with CUDA Python](#) issued by NVIDIA
- [Advanced Visualization with PyMol 3](#) issued by Schrödinger
- [Computational Chemistry School](#) at Novartis Cambridge, MA
- [Protein-ligand Gold Docking](#) issued by Cambridge Crystallographic Data Centre (CCDC)
- [Coleman Leadership Program](#) issued by Cornell Engineering and SLOAN

Research Experience

Massachusetts Institute of Technology (MIT) – Cambridge, MA Sept 2020 – Current

Co-Advisors: Dr. Forest White and Dr. Heather J. Kulik

- Graduate student researching how non-covalent interactions drive substrate positioning
- 24 peer-reviewed manuscripts on small molecule-protein binding, enzymes, and simulation.
- Software lead developer: [QuantumPDB](#), [PyEF](#), [QuantumAllostery](#), [CADDKit](#), [MolecuLearn](#)

Huntsman Cancer Institute (HCI) – Salt Lake City, UT Dec 2018 – Aug 2020

Oncological Sciences Department

- Full-time cancer computational and wet-lab biologist researching small cell lung cancer (SCLC)
- Gene expression bioinformatics and wet lab protocol optimization ([IHC](#))

National Institutes of Health (NIH) – Bethesda, MD Jul 2018 – Sept 2018

Advisor: Dr. Nico Tjandra

- Biophysics internship requiring electron microscopy, light scattering, NMR, circular dichroism

Dana-Farber/Harvard Cancer Center (DFCI/HCC) – Boston, MA Apr 2018 – Jul 2018

Advisor: Dr. Haribabu Arthanari

- Biophysics internship requiring NMR, protein expression, biochemical pathway engineering

Advisor: Dr. Steven L. Castle

- Organic synthesis and computationally modelling (QM/MM) of anticancer peptides
- Engineered a potent anticancer peptide in collaboration with *Bristol-Myers Squibb* (BMS).

Specific Software and Tools

- MD AMBER and OpenMM
- QM TeraChem and ORCA
- Schrödinger FEP+
- Schrödinger Maestro
- Schrödinger Glide
- Schrödinger LiveDesign
- AMBER MMGBSA/MMPBSA
- AutoDock Vina
- CCDC Gold docking
- Linux, Bash, SLURM, SGE
- PyMol, VMD, ChimeraX, Avogadro
- CREST conformer search

Skills and Specializations

- Spanish (bilingual)
- Computational chemistry
- Computer-Aided Drug Discovery (CADD)
- Structure activity relationships (SAR)
- End-point free energy calculations
- ADMET principles
- [Molecular visualization expert](#)
- Lead Python software developer ([GitHub](#))
- Molecular dynamics (MD) simulations
- Quantum mechanical (QM) calculations
- Density functional theory (DFT)
- Hybrid QM/MM multiscale simulations
- High-performance Computing
- Machine Learning
- Enhanced sampling (Metadynamics/TPS)
- High-performance computing (HPC)
- Molecular docking
- High-throughput virtual screening

Fellowships

MIT Whitehead Fellowship	2021
Vice Chancellor's Inclusive Excellence Fellowship	2020
National Science Foundation Graduate Research Fellowship (NSF GRFP)	2019
Tolero Pharmaceuticals Fellowship	2018
Inspired Learning University Fellowship	2018
Simmons Center for Cancer Research Fellowship (SCCR)	2017

Prestigious Awards

Siebel Scholars Award	2025
ACS Chemical Computing Group (CCG) Research Excellence Award	2024
Alfred P. Sloan Exemplary Mentorship Award (UCEM)	2019
Elva Pedersen Jorgenson Scholarship	2019
Warren Rollins and Murdell Hull Scholarship	2019
National Hispanic Fund Scholarship (HSF)	2018
Lieutenant Governor's Community Service Award	2018
National President's Volunteer Service Award	2018
National Honor Society Phi Kappa Phi Outstanding Student Award	2018
Karl G. Maeser Scholarship	2018
University Undergraduate Research Award (URA)	2018
National Barry Goldwater Scholarship	2017
National American Chemical Society Scholars Award (ACS)	2017
Full-Ride Merit-Based Scholarship	

2017

Front Cover Highlighted Publications ([Google Scholar](#)) ([ORCID](#)) ([GitHub](#))

1. [\[Front Cover\]](#) **Kastner, D.W.**; Nandy, A.; Mehmood, R.; Kulik, H.J. Mechanistic Insights into Substrate Positioning That Distinguishes Non-heme Fe(II)/ α -Ketoglutarate-Dependent Halogenases and Hydroxylases. *ACS Catal.* **2023**, 13 (4), 2489-2501. [DOI](#).
2. [\[Front Cover\]](#) Duan, C.; Nandy, A.; Terrones, G.G.; **Kastner, D.W.**; Kulik, H.J. Active Learning Exploration of Transition-Metal Complexes to Discover Method-Insensitive and Synthetically Accessible Chromophores. *JACS Au* **2023**, 3 (2), 391-401. [DOI](#).
3. [\[Front Cover\]](#) Del Rio Flores, A.; **Kastner, D.W.**; Du, T.; Narayanamoorthy, M.; Shen, Y.; Cai, W.; Vennelakanti, V.; Zill, N.A.; Dell, L.B.; Zhai, R.; Kulik, H.J.; Zhang, W. Probing the Mechanism of Isonitrile Formation by a Non-Heme Iron(II)-Dependent Oxidase/Decarboxylase. *JACS* **2022**, 144 (13), 5893-5901. [DOI](#).
4. [\[Front Cover\]](#) Nazemi, A.; Steeves, A.H.; **Kastner, D.W.**; Kulik, H.J. Influence of the Greater Protein Environment on the Electrostatic Potential in Metalloenzyme Active Sites: The Case of Formate Dehydrogenase. *J. Phys. Chem. B* **2022**, 126 (22), 4069-4079. [DOI](#).
5. [\[Front Cover\]](#) Lo, C.L.L.; Joaquin, D.; Moyá, D.A.; Ramos, A.; **Kastner, D.W.**; White, S.A.; Christensen, B.L.; Naglich, J.G.; Degnen, W.J.; Castle, S.L. Synthesis and Evaluation of Potent Yaku'amide A Analogs. *Chem. Sci.* **2022**, 13, 1899-1905. [DOI](#).

Other Peer-reviewed Publications ([Google Scholar](#)) ([ORCID](#)) ([GitHub](#))

6. Colin, Y.K.; **Kastner, D.W.**; Mitchell, J.A.; Gutierrez, M.A.; Yao, S.J.; Neumann, N.E.; Kulik, H.J.; Weng, J. Tracing the Stepwise Darwinian Evolution of a Plant Halogenase. *Science Advances*. **Accepted**.
7. Reinhardt, R.C.; **Kastner, D.W.**; Kulik, H.J. Role of Active Site Residues and Weak Noncovalent Interactions in Substrate Positioning in N,N-Dimethylformamidase. *ChemRxiv*. **2025**. [DOI](#).
8. Keys, A.M.; **Kastner, D.W.**; Kastner, D.W.; Kiessling, L.L.; Kulik, H.J. CH- π Interactions Confer Orientational Flexibility in Protein-Carbohydrate Binding Sites. *ChemRxiv*. **2025**. [DOI](#).
9. Keys, A.M.; **Kastner, D.W.**; Kiessling, L.L.; Kulik, H.J. The Energetic Landscape of CH- π Interactions in Protein-Carbohydrate Binding. *Chemical Science*. **2025**. [DOI](#).
10. Torrens-Spence, P.M.; Matos, O.J.; **Kastner, D.W.**; Li, T.; Glinkerman, M.C.; Sherk, J.; Wang, Y.; Kulik, H.J. Weng, J. Mechanistic Basis for the Emergence of EPS1 as a Catalyst in Plant Salicylic Acid Biosynthesis. *Nat. Commun.* **2024**. [DOI](#).
11. Del Rio Flores, A.; Zhai, R.; **Kastner, D.W.**; Seshadri, K.; Yang, S.; Matias, K.D.; Shen Y.; Cai, W.; Narayanamoorthy, M.; Do, N.B.; Xue, Z.; Marzooqi, D.A.; Kulik, H.J.; Zhang, W. Enzymatic Synthesis of Azide by a Promiscuous N-nitrosylase. *Nature Chemistry*. **2024**. [DOI](#).
12. Kevlishvili, I.; Vakil, J.; **Kastner, D.W.**; Huang, X.; Craig, S.; Kulik, H. High-Throughput Discovery of Ferrocene Mechanophores with Enhanced Reactivity and Network Toughening. *ChemRxiv*. **2024**. [DOI](#).
13. **Kastner, D.W.**; Reinhardt, C.R.; Adamji, H.; Manetsch, M.; Roman-Leshkov, Y.; Kulik, H.J. Dynamic Charge Distribution as a Key Driver of Catalytic Reactivity in an Artificial Metalloenzyme. *ChemRxiv*. **2024**. [DOI](#).
14. Edholm, F.; Nandy, A.; Reinhardt, C.R.; **Kastner, D.W.**; Kulik, H.J. Protein3D: Enabling analysis and extraction of metal-containing sites from the Protein Data Bank with molSimplify. *J. Comput. Chem.* **2023**, 45 (6), 352-361. [DOI](#).
15. Kim, C.Y.; Mitchell, A.J.; **Kastner, D.W.**; Albright, C.E.; Michael, A.G.; Glinkerman, C.M.; Kulik, H.J.; Weng, J. Emergence of a protein exchange-based isomerization and lactonization mechanism in the plant coumarin synthase COSY. *Nat. Commun.* **2023**, 14 (597). [DOI](#).
16. Nandy, A.; Adamji, H.; **Kastner, D.W.**; Vennelakanti, V.; Nazemi, A.; Liu, M.; Kulik, H.J. Using Computational Chemistry to Reveal Nature's Blueprints for Single-Site Catalysis of C-H Activation. *ACS Catal.* **2022**, 12 (15), 9281-9306. [DOI](#).

17. Nandy, A.; Terrones, G.; Arunachalam, N.; Duan, C.; Duan, C.; **Kastner, D.W.**; Kulik, H.J. MOFSimplify, Machine Learning Models with Extracted Stability Data of Three Thousand Metal-Organic Frameworks. *Nat. Sci. Data* **2022**, 9 (74). [DOI](#).
18. Olsen, R.R.; Ireland, A.S.; **Kastner, D.W.**; Groves, S.M.; Spainhower, K.B.; Pozo, K.; Kelenis, D.P.; Whitney, C.P.; Guthrie, M.R.; Wait, S.J.; Soltero, D.; Witt, B.L.; Quaranta, V.; Johnson, J.E.; Oliver, T.G. ASCL1 Represses a SOX9+ Neural Crest Stem-like State in Small Cell Lung Cancer. *Genes Dev.* **2021**, 37, 13-14. [DOI](#).
19. Ireland, A.S.; Micinski, A.M.; **Kastner, D.W.**; Guo, B.; Wait, S.J.; Spainhower, K.B.; Conley, C.C.; Chen, O.S.; Guthrie, M.R.; Soltero, D.; Qiao, Y.; Huang, X.; Tarapcsak, S.; Devarakonda, S.; Chalishazar, M.D.; Gertz, J.; Moser, J.C.; Marth, G.; Puri, S.; Witt, B.L.; Spike, B.T.; Oliver, T.G. MYC Drives Temporal Evolution of Small Cell Lung Cancer Subtypes by Reprogramming Neuroendocrine Fate. *Cancer Cell* **2020**, 38 (1), 60-78. [DOI](#).
20. Joaquin, D.; Lee, M. A.; **Kastner, D. W.**; Singh, J.; Morrill, S. T.; Damstedt, G.; Castle, S. L. Impact of Dehydroamino Acids on the Structure and Stability of Incipient 3₁₀-Helical Peptides. *J. Org. Chem.* **2020**, 3 (85), 1601-1613. [DOI](#).
21. Morris, D.L.; **Kastner, D.W.**; Johnson, S.; Strub, M.; He, Y.; Bleck, C.K.; Lee, D.; Tjandra, N.; Humanin induces conformational changes in the apoptosis regulator BAX and sequesters it into fibers, preventing mitochondrial outer-membrane permeabilization. *J. Biol. Chem.* **2019**, 50 (294), 19055-19065. [DOI](#).
22. **Kastner, D.W.** Computational Modelling of Peptides Containing Non-Standard Amino Acids *Undergraduate Honors Thesis* **2019**. *ScholarsArchive*, 61. [ISSN](#).
23. **Kastner, D.W.** The Ultimate Triumph of Truth. *Brigham Young University Commencement*, **2019**. URL: speeches.byu.edu.
24. Ashraf, N.M.; Krishnagopal, A.; Hussain, A.; **Kastner, D.W.**; Sayed, A.M.; Mol Y.K.; Swaminathan. K.; Zeeshan, N. Engineering of serine protease for improved thermostability and catalytic activity using rational design. *Int. J. Biol. Macromol.* **2018**, 126, 229-236. [DOI](#).
25. Jalan, A.; **Kastner, D.W.**; Webber, K.G.I.; Smith, M.S.; Price, J.L.; Castle, S.L. Bulky dehydroamino acids enhance proteolytic stability and folding in β -hairpin peptides. *Org. Lett.* **2017**, 19 (19), 5190-5193. [DOI](#).
26. Ashraf, N.M.; Imran, K.; **Kastner, D.W.**; Ikram, K.; Mushtaq, A.; Hussain, A.; Zeeshan, N. Potential involvement of mi-RNA 574-3p in progression of prostate cancer: A bioinformatic study. *Mol. Cell. Probes* **2017**, 36, 21-28. [DOI](#).

Oral and Poster Presentations

1. **Kastner, D.W.**; Clorice, R.; Adamji, H.; Manetsch, M.; Kulik, H.J. Mechanistic Insights from Electronic Properties in Artificial Mini-enzymes. *ACS Spring Conference Denver*, **2024**. [DOI](#).
2. **Kastner, D.W.**; Nandy, A.; Mehmood, R.; Kulik, H.J. Substrate positioning in Non-heme Fe(II)/ α -ketoglutarate-dependent halogenases and hydroxylases: A computational study. *Virtual Conference on Chemistry and its Applications (VCCA)*, **2023**. [Video recording](#).
3. **Kastner, D.W.**; Jalan, A.; Castle, S. L. Conformational ensemble calculations of proteolytically stable β -hairpins containing bulky α,β -dehydroamino acids. *American Chemical Society 254th National Meeting*, Washington D.C., **2017**.
4. **Kastner, D.W.**; and Castle, Steven L., Progress toward synthetically simplified natural anticancer peptide (2018). *Library Undergraduate Poster Competition* **2018**, 5. [ISSN](#)
5. **Kastner, D.W.**; Castle, S. L. Computational predictions β -hairpins containing bulky dehydroamino acids. *Scholars Archive* **2017**, 4. [ISSN](#)
6. **Kastner, D.W.**; Lo, C. C. L.; Castle, S. L. Progress towards a synthetically simplified anticancer peptide. *Student Research Conference (SRC)*, Provo, **2018**.

7. **Kastner, D.W.**; Jalan, A.; Castle, S. L. QM/MM analysis of proteolytically stable β -hairpins. *Student Research Conference (SRC)*, Provo, **2017**.
8. **Kastner, D.W.**; Castle, S. L. ONIOM geometry optimization of bulky dehydroamino acids in β -hairpins. *IEEE Poster Session*, Provo, **2017**.

Advanced Coursework and Instruction

Massachusetts Institute of Technology

<i>Biological Engineering</i> • Fundamentals of Drug Development (20.201)	Grade: A (5.0)
<i>Biological Engineering</i> • Research Experience in Biopharma (20.930)	Grade: A (5.0)
<i>Biological Engineering</i> • Protein Engineering (20.535)	Grade: A (5.0)
<i>Biological Engineering</i> • Analysis of Biological Networks (20.440)	Grade: A (5.0)
<i>Computer Science</i> • Modeling with Machine Learning: Algorithm to Apps (6.C51)	Grade: A (5.0)
<i>Chemical Engineering</i> • Machine Learning for Molecular Engineering (10.C51)	Grade: A (5.0)
<i>Chemical Engineering</i> • Computational Chemistry (10.637)	Grade: A (5.0)

Brigham Young University

<i>Physiology</i> • Biophysics (PDBIO 568)	Grade: A (4.0)
<i>Physiology</i> • Advanced Physiology (PDBIO 362)	Grade: A (4.0)
<i>Biology</i> • Cell Biology (BIO 360)	Grade: A (4.0)
<i>Biology</i> • Computational Biology (BIO 362)	Grade: A (4.0)
<i>Chemistry</i> • Biophysical Chemistry (CHEM 468)	Grade: A (4.0)
<i>Chemistry</i> • Organic Chemistry (CHEM 351 & 352)	Grade: A (4.0)
<i>Chemistry</i> • Biochemistry (CHEM 481)	Grade: A (4.0)
<i>Physics</i> • Molecular Dynamics (PDBIO 550R)	Grade: A (4.0)
<i>Physics</i> • Electricity and Magnetism (PHSCS 220)	Grade: A (4.0)
<i>Physics</i> • Newtonian and Modern Physics (PHSCS 121 & 123)	Grade: A (4.0)

Teaching Experience

MIT Instructor • *Biological Engineering* • 3D Scientific Rendering (20.S947)

MIT Teaching Assistant • *Chemical Engineering* • Computational Chemistry (10.437)

MIT Teaching Assistant • *Biological Engineering* • Principles of Molecular Bioengineering (20.420)

BYU Teaching Assistant • *Cell Biology & Physiology* • Cell Biology (PDBIO 360)

Current Societal and Honors Affiliations

Foundation for Science and Disability (FSD)

Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS)

National Spanish Speaker Honor Society (SDP)

American Chemical Society (ACS) – Computers in Chemistry (COMP) Division

The Protein Society (TPS)

National Scientific Research Honor Society Sigma Xi

Press

[“The art of the enzyme”](#) MIT News, **2024**.

[“Computational model captures the elusive transition states of chemical reactions”](#) MIT News, **2023**.

[“Scientists use computational modeling to design “ultrastable” materials”](#) MIT News, **2023**.

[“Computational modeling guides development of new materials”](#) MIT News, **2022**.

[“The Ultimate Triumph of Truth”](#) BYU Speeches, **2019**.

[“Part of the search: Q&A with BYU Commencement Speaker”](#) BYU Honors Program News, **2019**.

["A wonderful world to share joy, show love, create connections"](#) *BYU News*, **2019**.

["Kastner chosen as commencement speaker at BYU"](#) *Napa Valley Register*, **2019**.

["Rex Lee Run unites those affected and impacted by cancer"](#) *Daily Herald*, **2018**.

["Kastner receives fellowship for cancer research"](#) *Napa Valley Register*, **2018**.

["Napa's Kastner family creates artwork for new dealership"](#) *Napa Valley Register*, **2016**.